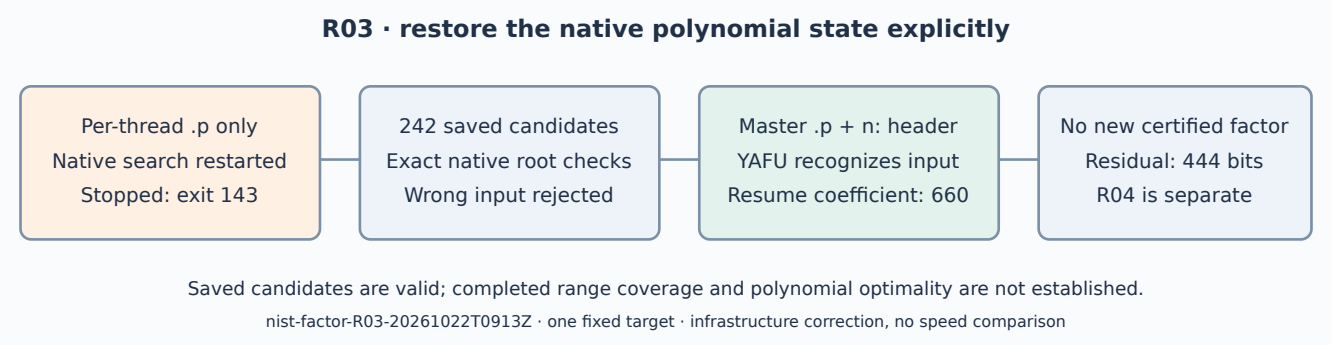
R03: a polynomial restore failure and its correction

Run nist-factor-R03-20261022T0913Z, started 2026-10-22 09:13:55 UTC. The user requested continued factorization. One P-256 native NFS attempt was launched from R02's per-thread checkpoint under the separately frozen protocol, native limit 1800 seconds, outer 1860 seconds plus 30 seconds kill grace, seed 20261022 and two threads. Pinned YAFU: c7d90283972820f99b3b4920bfc1eff05c1f6342; PARI/GP 2.17.2.



Only one fixed target was attempted; an infrastructure failure prevented new coverage from being established. The explanatory diagram replaces a comparative runtime graph. The 242 count is recorded by the native exact validator, not a count of tested coefficient ranges or new independent polynomials found in R03. No sampling interval or speed claim applies.

What actually happened

The restored files contained nfs.dat.0.p, while YAFU's native restore path requires master nfs.dat.p with an n: header. Its source at factor/nfs/nfs_filemanip.c checks that master file and otherwise selects NFS_STATE_STARTNEW; factor/nfs/nfs_poly.c normally merges a thread's file only after the batch returns. The interrupted R02 batch had not performed that merge. R03 therefore began at the old starting coefficient and repeated the first full candidate. A byte comparison confirms that repeat; this does not establish that every produced record is a duplicate.

SIGINT was recorded but did not stop the process. SIGTERM then ended it with exit 143; exact timestamps and reasons are in raw manual-{interrupt,terminate}.json. No native factor hint was returned. Independent PARI refinement/replay recovered the unchanged verified partial partition. The 512-bit whole target still has a 444-bit composite residual and largest currently proven prime factor of 33 bits. Eight of nine whole targets remain complete, resolving thirteen of fourteen original fields.

Recovery and controls

The additive recovery protocol preregistered exact checks before recovery. Native GP validated all 242 complete saved degree-5 candidates from R02: coefficient syntax, nonzero leading term, rational pair, invertibility of Y1, and zero of the algebraic polynomial at -Y0/Y1 modulo the literal input. It wrote the original candidate blocks with the correct input header and no invented timing. The first parser version mishandled the string delimiter; its error and empty proposal are retained. The corrected parser passes.

A different literal input fails the exact root check. The independent native YAFU recognition diagnostic on a separate copy, bounded to three seconds plus one second grace, prints poly file matches input and last leading coefficient was 660; exit 124 is the diagnostic bound, not a production negative. Any incidental arithmetic is setup work. The portable checkpoint restores with the same input and independently replays the prime certificates and exact whole-target product. Neither recognition nor partial replay proves complete factorization.

The resume wrapper now refuses unrecognized per-thread-only state. A configurable native -pbatch records the selected batch; R04 separately freezes a 120-coefficient batch to permit earlier durable checkpoint merging. That future search is not counted as R03 coverage.

Evidence, limits and remaining work

Raw runs.tar.xz preserves both native attempts' inherited history, the intervention logs, original/repeated candidates, parser failure, recovery verdicts, wrong-input rejection and recognition diagnostic. Check its SHA-256 before extracting. The loose master file and derived checkpoint excerpt are named directly by the replay/job and remain loose under the AGENTS.md section 13a exception. The full raw archive is distinct.

The host has four CPUs and 16 GiB RAM, L0 isolation. Python only renders this document. Existing R02 product proofs and prime certificates still apply to the unchanged partition; no new factor identity is claimed. The remaining factorization obstruction and separate bulk-publication access gap remain explicit. Historical generators, upstream trait defects and the historical skip reason are unresolved. No finite failed search proves impossibility or a factor-size lower bound.

The local diagram and PDF are new; IC scoreboards, timelines, performance figures and seed-study diagrams contain no affected result and are unchanged.